

Team Name : Hyperion

Team Leader Name : Anuj Selvakumar

Problem Statement : AI-Powered Digital Twin of India's Climate using India's National Data

Team Members

Team Leader

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How is Hyperion Different?

Opportunity · 1 / 3

Current climate platforms answer	Hyperion Climate Twin answers
“What will happen?”	*“What should we do next?”*

Unlike existing forecasting systems or digital dashboards that only visualize climate conditions, **Hyperion continuously builds a living Digital Twin of India's climate** using national datasets, AI foundation models, physics-informed reasoning, and autonomous AI agents. Users can interactively simulate future scenarios, compare policy alternatives, and receive optimized recommendations through reinforcement learning—transforming climate prediction into intelligent climate decision-making.

Potential reach: 1.4 billion people covered · 60M+ farmers dependent on monsoon data · 48-hour advance warning improvement with AI-driven flood and heatwave agents

How will Hyperion Solve the Problem?

Opportunity · 2 / 3

Hyperion solves the problem by creating a continuously evolving virtual replica of India's climate that combines national datasets with AI foundation models. Real-time observations are assimilated into a Hybrid Digital Twin, where AI predicts future climate conditions and physics-informed validation ensures reliable simulations. Specialized AI agents analyze impacts across floods, heatwaves, agriculture, reservoirs, urban infrastructure, and air quality. Through interactive “What-if” simulations and reinforcement learning, decision-makers can evaluate multiple adaptation strategies before implementation, enabling proactive climate resilience instead of reactive disaster management.

Hackathon Prototype Scope: The 30-hour finale prototype will demonstrate end-to-end data ingestion from ERA5 and IMD APIs, Prithvi WxC inference for 72-hour forecasts, a live CesiumJS 3D dashboard with at least two domain agents (Flood and Heatwave), and one What-if simulation. Full multi-agent and RL integration is the post-hackathon roadmap.

Domain Intelligence Agents

- Floods
- Cyclone
- Heatwaves
- Agriculture
- Reservoirs
- Urban
- Air Quality

USP of the Proposed Solution

Opportunity · 3 / 3 — Six core differentiators

1

AI-powered Hybrid Digital Twin

Combines Foundation AI + Physics + Simulation + National datasets — not just one approach.

2

Multi-Agent Climate Intelligence

Specialized agents for floods, cyclones, agriculture, heatwaves, reservoirs, air quality & disaster response — that collaborate.

3

Interactive 4D Climate Simulation

Simulate +2°C warming, urban expansion, afforestation, reservoir operations, emission cuts & rainfall changes — visualize outcomes instantly.

4

Reinforcement Learning Decision Engine

Doesn't just predict — recommends the optimal adaptation strategy.

5

Explainable AI

Every recommendation includes confidence, reasoning, affected regions & expected impact.

6

Natural Language Climate Copilot

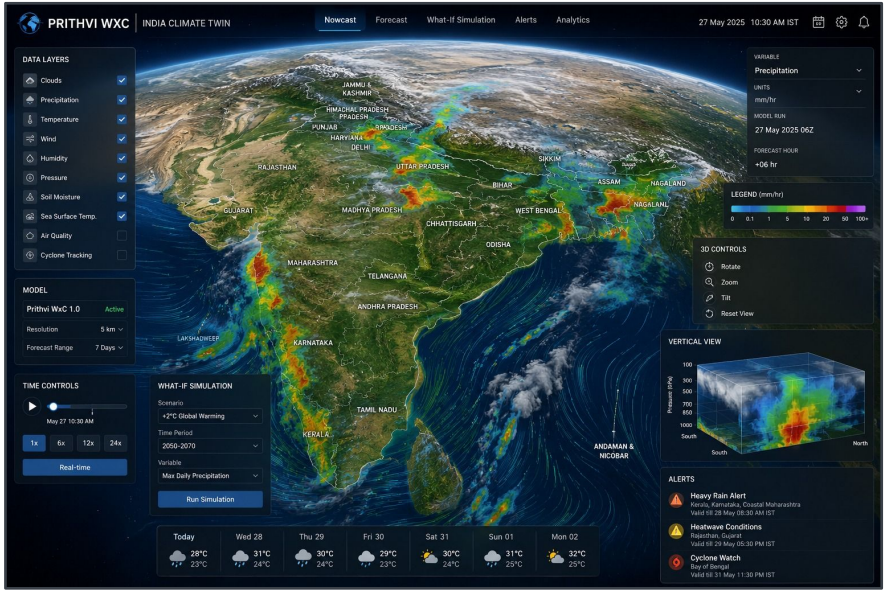
Just ask — “What happens if rainfall drops 20% in Karnataka?” — no complex GIS software needed.

Features Offered by the Solution

1 / 2 — Core platform & climate intelligence

Core Platform

- National Climate Data Integration (INSAT, MOSDAC, Oceansat, Sentinel, IMD) + ERA5 global reanalysis (ECMWF)
- AI Climate Forecasting via Prithvi WxC Foundation Model
- Physics-informed Forecast Validation
- India Climate Digital Twin with real-time state updates
- Interactive 3D Visualization — State ⟶ District ⟶ City
- Historical Climate Replay & Future Forecast



Live 3D India Climate Digital-Twin Dashboard (Prithvi WxC)

Features Offered by the Solution

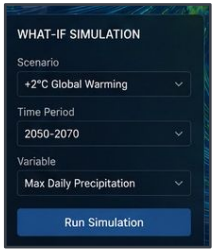
2 / 2 — Climate intelligence & decision support

Climate Intelligence

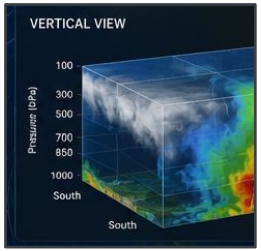
- What-if Scenario Simulation
- Flood Prediction & Early Warning
- Cyclone Tracking & Landfall Estimation
- Heatwave Prediction & Health Risk Alerts
- Reservoir Monitoring & Water-Level Forecast
- Air Quality Analysis & PM2.5 / AQI Mapping

Decision Support

- Agriculture Intelligence (crop stress, rainfall anomalies)
- RL-based Decision Recommendations for policy makers
- AI Climate Copilot (natural-language Q&A)
- Policy Impact Analysis — simulate before deploying



What-if Simulation



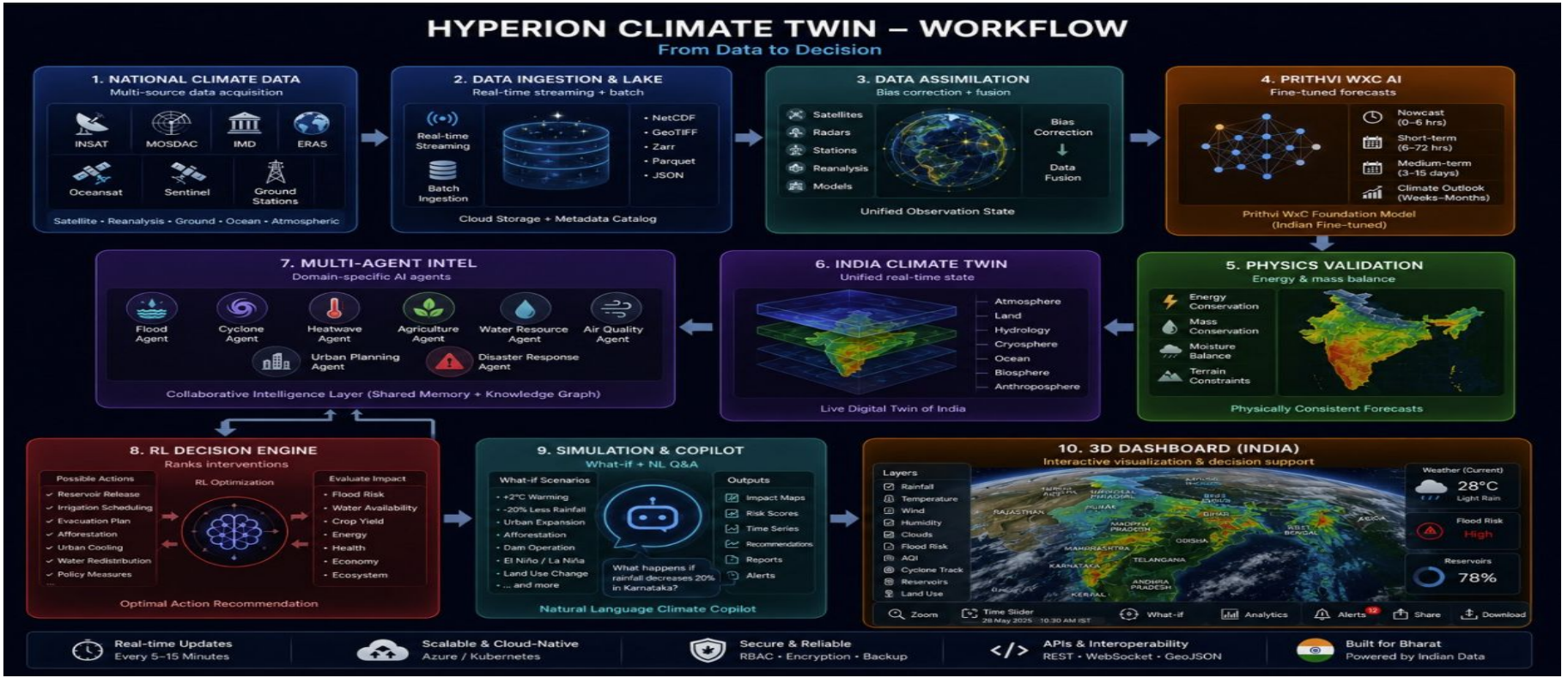
4D Vertical Profiles



Real-time Risk Alerts

Process Flow Diagram

End-to-end climate intelligence pipeline



UI Wireframe — Home Dashboard

Key UI Components

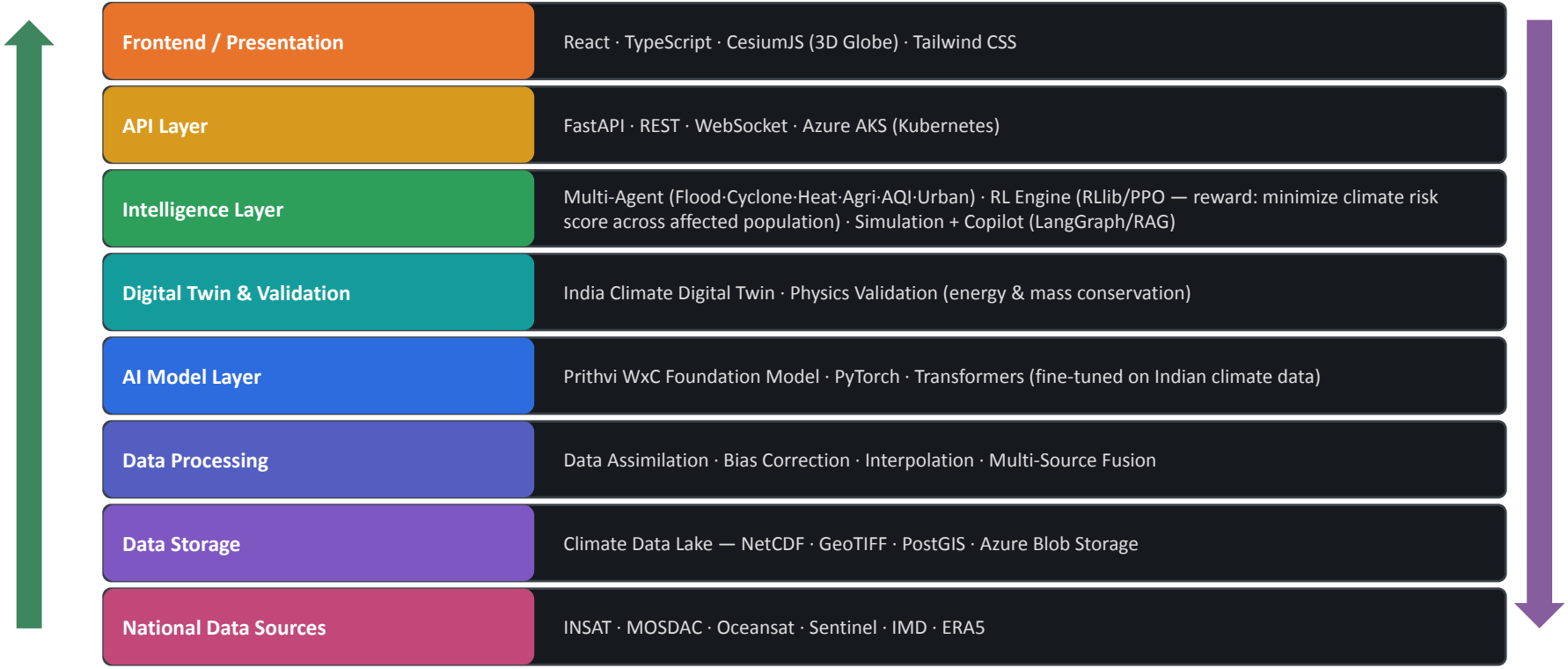
- Navigation Bar: Hyperion Climate Twin logo | Dashboard | Simulation | Agents | Copilot | Reports
- 3D India Globe (CesiumJS / WebGL): zoomable map with real-time climate overlays
- Weather Layers Panel: Temperature | Wind Speed | Rainfall | Air Quality | Sea Surface Temp
- Simulation Panel: Scenario selector (e.g. +2°C Warming) | Time Horizon | Run Simulation button
- Time Slider: Scrub between 1990 → Present → 2050 for historical replay & future forecast
- AI Climate Copilot Chat: type natural language questions, get instant AI-powered answers
- Agent Status Cards: Live status of Flood | Cyclone | Heatwave | Agriculture agents
- Alert Banner: Real-time risk alerts with severity level & recommended actions

Navigation Flow

- India Map View → Click State → District Dashboard → City-level Detail
- Simulation Mode → Select Variables → Run Model → View Impact Layers
- Copilot Mode → Ask Question → View AI Answer + Source Data + Map Highlight

Architecture Diagram

Layered system architecture (frontend → national data)



Technologies Used in the Solution

1 / 2 — Frontend, Backend & AI / ML

Frontend

- React
- TypeScript
- CesiumJS (3D globe)
- Tailwind CSS

Backend

- Python
- FastAPI
- PostGIS (geospatial DB)
- Redis (caching)

AI / ML

- Prithvi WxC (NASA/IBM)
- PyTorch
- HuggingFace Transformers
- LangGraph (agents)
- RLlib (reinforcement learning)

Technologies Used in the Solution

2 / 2 — Cloud, infrastructure & data

Cloud & Infrastructure

- Microsoft Azure
- Azure AI Foundry
- Azure Blob Storage
- AKS (Kubernetes)

Data Formats & Sources

- NetCDF (climate data)
- GeoTIFF (satellite imagery)
- Sentinel-2 / Sentinel-3 (ESA satellite imagery)
- ERA5 (ECMWF reanalysis)
- INSAT / MOSDAC / Oceansat (ISRO)
- IMD observations

Estimated Implementation Cost

Hackathon Prototype Cost

Component	Cost
National Datasets (INSAT, MOSDAC, Oceansat, ERA5)	₹0 — open / public access
Prithvi WxC Foundation Model	₹0 — open-source (NASA/IBM, Hugging Face)
Open-Source Libraries (PyTorch, LangGraph, RLlib, FastAPI)	₹0
CesiumJS (3D globe visualization)	₹0 — open-source
Development Effort	₹0
Model Fine-Tuning (Vultr A100/H100 GPU, 80 GB VRAM)	₹0 — $250 free credits for new accounts
Hackathon Deployment (Azure AKS / Blob / AI Foundry)	₹0 — $200 Azure free credit (1 month)
TOTAL Prototype Cost	₹0


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2026
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